

DESIGN AND FABRICATION OF AUTOMATIC WASTE SEGREGATION ROVER SYSTEM

A PROTOTYPE LAB REPORT

Submitted by

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In partial fulfillment for the award of the degree

of

BACHELOR OF ENGINEERING

IN

MECHATRONICS ENGINEERING

OCTOBER 2025

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SRI KRISHNA COLLEGE OF ENGINEERING AND TECHNOLOGY

**An Autonomous Institution | Approved by AICTE | Affiliated to Anna University | Accredited by NAAC with A++ Grade
Kuniamuthur, Coimbatore – 641008.**

OCTOBER 2025

SUSTAINABLE DEVELOPMENT GOALS

The Sustainable Development Goals are a collection of 17 global goals designed to blue print to achieve a better and more sustainable future for all. The SDGs, set in 2015 by the United Nations General Assembly and intended to be achieved by the year 2030, In 2015, 195 nations agreed as a blue print that they can change the world for the better. The project is based on one of the 17 goals.

Questions	Answer Samples
What is the main software function of the Automatic Waste Segregation Rover?	This project supports SDGs 7, 9, 11,12,and 15 with a microcontroller (Arduino/ESP32/Raspberry Pi), process sensor data in real-time, and control actuators such as motors, servo arms, and rover wheels. The software ensures correct waste detection, segregation, and mobility..
How does this project align with SDG 7: Affordable and Clean Energy ?	The rover is powered by rechargeable lithium-ion batteries and can be coupled with solar panels for charging. The software includes energy-efficient motor control algorithms like PWM (Pulse Width Modulation) to minimize power consumption..
How does the rover system support SDG 9: Industry, Innovation, and Infrastructure ?	The rover demonstrates automation and robotics in waste management, an innovative solution for smart cities. The software can be extended to cloud/IoT platforms for real-time monitoring, predictive maintenance, and AI-driven optimization.
How does the rover software contribute to SDG 15: Life on Land ?	Proper segregation reduces soil pollution and landfill overflow, protecting terrestrial ecosystems

BONAFIDE CERTIFICATE

Certified that this project report **DESIGN AND FABRICATION OF AUTOMATIC WASTE SEGREGATION ROVER SYSTEM** is the bonafide work of **NAVEEN RAGAVENDRA B (727723EUMT085), RISHVANTH J M (727723EUMT107), SABAREESH SUNDAR S S (727723EUMT110), SELVENDRAN S (727723EUMT123).**

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INTERNAL EXAMINER

EXTERNAL EXAMINER

ACKNOWLEDGEMENT

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We wish to convey our gratitude to **Dr. K . Porkumaran**, Principal for supporting us to do our project and offering adequate duration to complete our project.

We would like to express our grateful thanks to **Dr. M. Lydia**, Head of the Department, Department of Mechatronics Engineering for her encouragement and valuable guidance on this project.

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ABSTRACT

Rapid urbanization has intensified waste management challenges, with traditional methods often leading to overflowing bins and unhygienic surroundings. To overcome these issues, this project introduces Garbot, an IoT-enabled smart garbage bin that monitors waste levels in real-time and automates collection alerts. It minimizes manual intervention while promoting efficient and sustainable waste disposal.

Garbot features segregated compartments for wet, plastic, paper, and metal waste, ensuring disposal at the source. Equipped with ultrasonic sensors for fill-level detection and additional sensors for lid and material monitoring, the system enhances hygiene and efficiency. Powered by a rechargeable supply, it supports eco-friendly operation. Its mobile app-controlled mobility allows users to move the bin conveniently, improving accessibility and cleanliness.

Technically, Garbot integrates an Arduino UNO, ESP32, and sensor modules to track and transmit waste data. Mecanum wheels enable omnidirectional movement, while optimized design calculations ensure lightweight and efficient performance. The total cost of ₹5520 makes it economically viable for residential and public applications.

Overall, Garbot combines IoT, mechatronics, and sustainability to provide an effective waste management solution adaptable to smart city infrastructures. The control logic ensures that each type of waste is guided to the correct compartment—wet waste to the biodegradable bin, dry waste to the recyclable bin, and metallic waste to the scrap collection bin.

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CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

The growing concern of waste management has become a significant challenge in modern society due to rapid urbanization, industrialization, and population growth. Inefficient handling of waste often leads to overflowing bins, foul odor, environmental pollution, and health hazards. Traditional methods rely heavily on manual monitoring and collection, which not only increases operational costs but also demands extensive manpower.

To address these challenges, the project Garbot has been conceptualized and developed. Garbot is an IoT-enabled smart garbage bin designed to simplify waste segregation, monitoring, and disposal. The system incorporates multiple compartments for wet, dry, plastic, paper, and metal waste, ensuring proper segregation at the source. With the integration of ultrasonic sensors, proximity

detectors, and load cells, the bin continuously tracks waste levels and alerts users when it is near capacity. Furthermore, the rover is equipped with mecanum wheels and controlled through a mobile application, allowing users to move it conveniently to their location for waste disposal.

Garbot stands out as a sustainable and eco-friendly innovation by combining IoT technology with mechatronics engineering. The rechargeable power supply ensures reduced energy consumption, while the real-time monitoring feature eliminates the inefficiencies of manual supervision. By minimizing human contact with waste and providing automated alerts for collection. The project not only benefits households but also holds strong potential for application in public spaces, institutions, and municipalities. Ultimately, Garbot represents a step toward building smart, clean, and sustainable cities for the future.

1.2 OBJECTIVE OF PROJECT WORK

The objective of this project is to design and fabricate an automatic waste segregation rover that efficiently separates waste into different categories such as wet, dry, and metallic wastes. The system aims to reduce human intervention in waste segregation, improve the efficiency of waste management processes, and minimize environmental pollution by ensuring proper sorting and disposal of waste materials. The rover will utilize sensors to detect and categorize the waste and mechanisms to direct the waste to the appropriate bins for further processing or recycling. This automatic segregation system focuses on providing a cost-effective, reliable, and user-friendly solution to enhance waste handling and promote sustainable waste management practices.

This project not only emphasizes automation and accuracy in waste segregation but also contributes to the broader goal of sustainable waste management. By minimizing

manual handling, it helps in reducing health risks associated with waste collection and enhances the efficiency of municipal waste processing systems. The implementation of such an autonomous system can be highly beneficial in urban areas, industries, and public places where large quantities of waste are generated daily.

The system's scalability allows future upgrades such as IoT-based monitoring, data analytics for waste management, and integration with renewable energy sources like solar panels for power efficiency.

1.4 ORGANISATION OF CHAPTER

Figure 1.1 depicts the organization of chapters for the proposed prototype Lab.

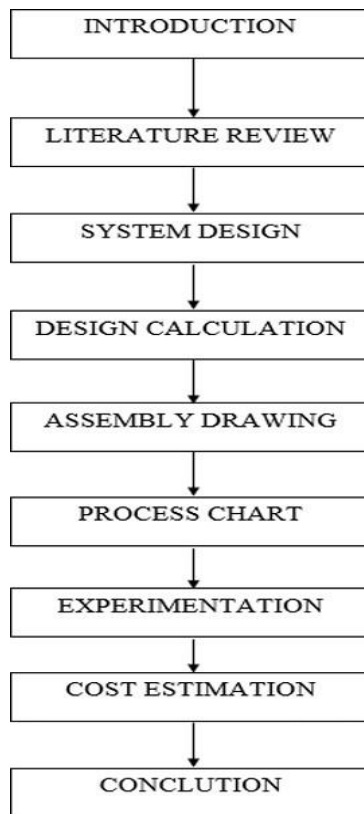


Fig no 1.1 Organisation of chapter

-1.3.1 Problem definition

The inefficient and often manual segregation of waste poses a significant environmental and public health challenge globally. Current waste management practices frequently rely on waste being mixed together at the collection point, which severely complicates and slows down the subsequent process of recycling and disposal. Improper segregation leads to a substantial portion of recyclable materials ending up in landfills, wasting valuable resources, increasing the volume of non-biodegradable waste, and contributing to soil and water contamination. Furthermore, manual sorting is a labor-intensive, hazardous, and undignified task for workers, exposing them to pathogens and toxic materials.

The core problem lies in the lack of a scalable, automated, and accurate pre-collection waste sorting system capable of handling different waste streams (e.g., wet, dry, metallic) directly at the source or an intermediate point. Existing automated systems are often large-scale, fixed industrial units, making them unsuitable for deployment in diverse, smaller-scale environments like residential areas, schools, or offices. There is a critical need for a smart, flexible, and mobile solution that can efficiently identify and separate common household and commercial waste types. This failure in initial segregation leads to higher operational costs for municipalities and private waste management companies due to the complexities of processing mixed waste, and ultimately undermines efforts toward achieving a true circular economy. Therefore, the goal is to develop an Automatic Waste Segregation Rover System to address these deficiencies by providing an autonomous, mobile, and accurate sorting capability.

1.3.2 System Design

This stage focuses on the overall architecture and functional design of Garbot-S. The system is built around separate compartments for biodegradable, plastic, paper, and metal waste. An Arduino and ESP32 microcontroller are used to process sensor signals and manage communication. The design integrates ultrasonic

sensors for fill detection, load cells for weight measurement, and proximity sensors for segregation. The rover mobility system, powered by mecanum wheels, allows the bin to be positioned conveniently for users. A mobile application is included for real-time monitoring, navigation control, and waste collection alerts, making the design user-friendly and efficient.

1.3.3 Design calculation

This section covers technical calculations for ensuring efficient performance:

- Load Distribution: 4 kg total weight → 1 kg per wheel.
- Wheel Selection: 60 mm mecanum wheels chosen for omnidirectional movement.
- Motor Speed: ~443 RPM for a speed of 5 km/h.
- Torque Requirement: 1.17 Nm total (sufficient for waste handling).
- Power Consumption: ~13.6 W for all four wheels.

These calculations ensure Garbot achieves reliable mobility and efficient operation while maintaining low energy consumption.

1.3.4 Assembly drawing

The assembly drawing section includes 2D/3D models of Garbot-S, showing:

- Placement of waste bins and segregation compartments.
- Position of sensors for accurate detection.
- Mechanical structure with frame, wheels, and motor layout.
- Electronic integration of Arduino, ESP32, and power system.
- Mobile app connection points.

These visuals guide the construction and ensure proper integration of mechanical and electronic components.

1.3.5 Process chart

The process chart outlines how Garbot-S operates step by step:

- Waste is deposited into the bin.
- Sensors (ultrasonic, moisture, proximity, load cells) detect type and fill level.
- The system classifies waste into wet, dry, plastic, or metal bins.
- The IoT module (ESP32) sends real-time data to the mobile app.
- Alerts are generated when bins approach capacity or need liner replacement.
- The rover can be navigated remotely for collection.

This ensures a clear understanding of the workflow and helps in troubleshooting.

1.3.6 Experimentation

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- The rover can be navigated remotely for collection.

This ensures a clear understanding of the workflow and helps in troubleshooting.

1.3.7 Cost estimation

Cost estimation involves listing and analyzing all project expenses:

- Components: Sensors, Arduino, ESP32, motors, wheels, frame, and batteries.
- Fabrication Costs: Mechanical frame, wiring, and assembly.
- Maintenance Costs: Replacement of sensors, batteries, or liners over time.

This section ensures that Garbot is cost-effective, scalable, and economically viable for public and residential use.

1.3.8 Summary

Garbot is an automated waste segregation rover designed to address the challenges of manual and inefficient waste management. It uses Arduino and ESP32 microcontrollers with sensors such as ultrasonic, proximity, moisture, and load cells to detect and

classify waste into biodegradable, plastic, paper, and metal categories. The mecanum wheel system ensures smooth mobility, while a mobile app enables real-time monitoring, navigation, and collection alerts. Design calculations ensure energy-efficient performance, and cost estimation confirms affordability. Garbot-S provides a scalable, smart, and sustainable solution for effective waste segregation in residential, institutional, and public environments.

CHAPTER 2

LITERATURE REVIEW

The literature on smart waste management systems highlights key aspects such as waste segregation technologies, IoT-based monitoring, automation methods, and sustainability benefits. Researchers have investigated various sensor technologies, including ultrasonic sensors for fill-level detection, load cells for weight measurement, and moisture and metal sensors for effective segregation. In addition, advancements in communication modules such as GSM, Wi-Fi, and ESP32-based IoT frameworks have been enable real-time monitoring and alerts for efficient waste collection.

A review of previous studies also provides insight into challenges such as ensuring accurate waste classification, reducing human involvement in hazardous waste handling, improving energy efficiency, and enhancing system scalability for urban environments. The integration of machine learning approaches like CNNs and YOLO for image-based waste classification, along with renewable energy options, further emphasizes the role of smart bins in promoting cleaner and more sustainable cities. This literature review seeks to synthesize existing research,

In summary, the literature review underscores the diverse and multidisciplinary nature of smart waste management. It highlights the significant advancements in sensor technologies, IoT integration, machine learning, and mobile-based monitoring that have contributed to the development of more efficient, reliable, and eco-friendly systems.

2.1 LITERATURE SURVEY :

1.Abhishek Singh, A. Banerjee, S. Chatterjee, H.K. Das, N.K. Das, Sirshendu Saha, & Hari D. Mandal (2024)

This paper presents a sustainable smart trash bin system that employs MQ137, TGS2611, and metal sensors for waste detection. The system segregates biodegradable, plastic, metal, glass, and mixed waste at the source, thereby enhancing recycling efficiency and reducing landfill burden.

2.Rubina Dutta, Jeetasha Jeetasha, Harsimran Jit Kaur, Deepti Prit Kaur, & Priyanka Malhotra (2023)

The authors developed “Sort-O-Matic,” an efficient waste segregation system that utilizes metal proximity, moisture, IR, and ultrasonic sensors. Their work demonstrates reduced human contact with hazardous waste and improved separation efficiency across multiple waste streams.

3.V. Lavanya, G. Abikash Shivam, S. Shanmugasundaram, & R. Deepalakshmi (2023)

This study introduces a CNN-based smart waste segregation and collection system that leverages YOLOv8 and deep learning for real-time waste classification. The system categorizes waste into biodegradable, recyclable, and non-recyclable classes, with potential scalability to large-scale urban applications.

4.Atharv Dahale (2022)

This work combines IoT-enabled fill-level monitoring with CNN-based waste classification. By supporting smart scheduling of waste collection trucks, the system effectively prevents overflow and improves recycling throughput in urban waste management.

5.Bhupinder Singh, Christian Kaunert, & Pankaj Kumar Tyagi (2021)

This research focuses on sustainable waste segregation and collection protocols in rural tourism areas, employing industrial IoT-based systems. The study emphasizes low-power operation and reducing waste leakage into natural landscapes, showing the applicability of smart segregation beyond urban environments.

6.Kale & Patil (2018)

Their study proposes the development of a low-cost automatic waste segregator using microcontroller-based design. The system detects and separates metal, wet, and dry waste with IR and moisture sensors while including GSM-based monitoring for timely collection, demonstrating an affordable solution for smaller communities.

7.Rathod & Shukla Singh (2018)

This work focuses on the design of an intelligent waste segregator for smart cities using Arduino-based control systems. It integrates real-time alerts to collection units and highlights modular, scalable designs that can be adapted to city-wide deployment.

CHAPTER 3

DESIGN CALCULATION

Garbot is an IoT-enabled smart waste segregation system designed to promote sustainable waste management. It integrates multiple sensors to identify and segregate wet, dry, plastic, and metallic waste into dedicated bins. Real-time fill level monitoring with ultrasonic sensors ensures timely alerts via a mobile application, while mobility features allow the rover to reach users conveniently. Powered by a rechargeable battery, Garbot reduces human effort, enhances hygiene, and supports smart city waste management initiatives.

The **design and calculation** process ensures the system is functional and reliable. Key aspects include:

- Load, torque, power, and battery calculations ensure Garbot's structure, mobility, and sensors operate efficiently under real conditions.

3.1 Power Requirement Calculation

- **Microcontroller (Arduino Uno Rev3 SMD):** $\sim 50 \text{ mA @ } 5\text{V} \rightarrow \mathbf{0.25 \text{ W}}$
- **Ultrasonic Sensor (HC-SR04):** $\sim 15 \text{ mA @ } 5\text{V} \rightarrow \mathbf{0.075 \text{ W}}$
- **IR Sensors (x3):** $\sim 20 \text{ mA each @ } 5\text{V} \rightarrow 3 \times 0.1 \text{ W} = \mathbf{0.3 \text{ W}}$
- **Servo Motors (x2 for flap movement):** $\sim 500 \text{ mA peak @ } 5\text{V} \rightarrow 2 \times 2.5 \text{ W} = \mathbf{5 \text{ W}}$
- **DC Motor/Stepper Motor (for conveyor/rotating plate):** $\sim 1 \text{ A @ } 12\text{V} \rightarrow \mathbf{12 \text{ W}}$
- **Wi-Fi/IoT Module (ESP8266/ESP32 if used):** $\sim 250 \text{ mA @ } 3.3\text{V} \rightarrow \mathbf{0.8 \text{ W}}$

Total Load $\approx 18.4 \text{ W}$ (round off 20 W)

3.2 Bin Size Calculation

Let's assume daily household waste $\approx \mathbf{2\text{--}3 \text{ kg/day}}$.

- **Wet Waste Density (biodegradable):** $\sim 400\text{--}500 \text{ kg/m}^3$
- **Dry Waste Density (paper/plastic):** $\sim 100\text{--}150 \text{ kg/m}^3$
- **Metal Waste Density:** $\sim 2500\text{--}3000 \text{ kg/m}^3$

If 2.5 kg waste/day $\rightarrow$ assume **50% wet, 40% dry, 10% metal**:

- Wet Waste Volume = $1.25 \text{ kg} \div 500 = \mathbf{0.0025 \text{ m}^3 (2.5 \text{ L})}$
- Dry Waste Volume = $1.0 \text{ kg} \div 120 = \mathbf{0.0083 \text{ m}^3 (8.3 \text{ L})}$
- Metal Waste Volume = $0.25 \text{ kg} \div 2700 \approx \mathbf{0.0001 \text{ m}^3 (0.1 \text{ L})}$

Bin Capacity (with $2\times$ safety factor):

- Wet Bin: **5 L**
- Dry Bin: **15 L**
- Metal Bin: **1 L**

3.3 Motor Torque Calculation (for flap/rotating mechanism)

flap of **0.15 m length** and waste load $\sim 0.5 \text{ kg}$.

- Torque (T) = Force $\times$ Distance
- Force = $m \times g = 0.5 \times 9.81 = \mathbf{4.9 \text{ N}}$
- Distance = **0.15 m**
- $T = 4.9 \times 0.15 \approx 0.735 \text{ Nm}$

Servo motor with torque rating $\geq \mathbf{1.5 \text{ kg}\cdot\text{cm} (0.147 \text{ Nm})}$ won't be enough.
Use standard servo (**10–15 kg·cm, i.e., 1–1.5 Nm**) for reliable operation.

3.4 Sensor Placement Calculation

- **Ultrasonic Sensor:** Placed at $\sim 10 \text{ cm}$ above bin inlet; detects waste presence if $< 8 \text{ cm}$.
- **IR Sensor:** Detects material reflection for dry/wet classification at $\sim 2\text{--}3 \text{ cm}$.
- **Inductive Proximity Sensor:** Detects metallic objects within $2\text{--}5 \text{ cm}$ range.

3.5 Battery Runtime Calculation

For a **12V 2200 mAh Li-ion battery** (capacity = 26.4 Wh):

- System load $\approx 20 \text{ W}$
- Runtime = $26.4 \div 20 = \mathbf{\sim 1.3 \text{ hours}}$

CHAPTER 4

SYSTEM ARCHITECTURE

4.1 DESIGNING AND FABRICATION OF AUTOMATIC WASTE SEGREGATION ROVER SYSTEM.

4.1Side view:

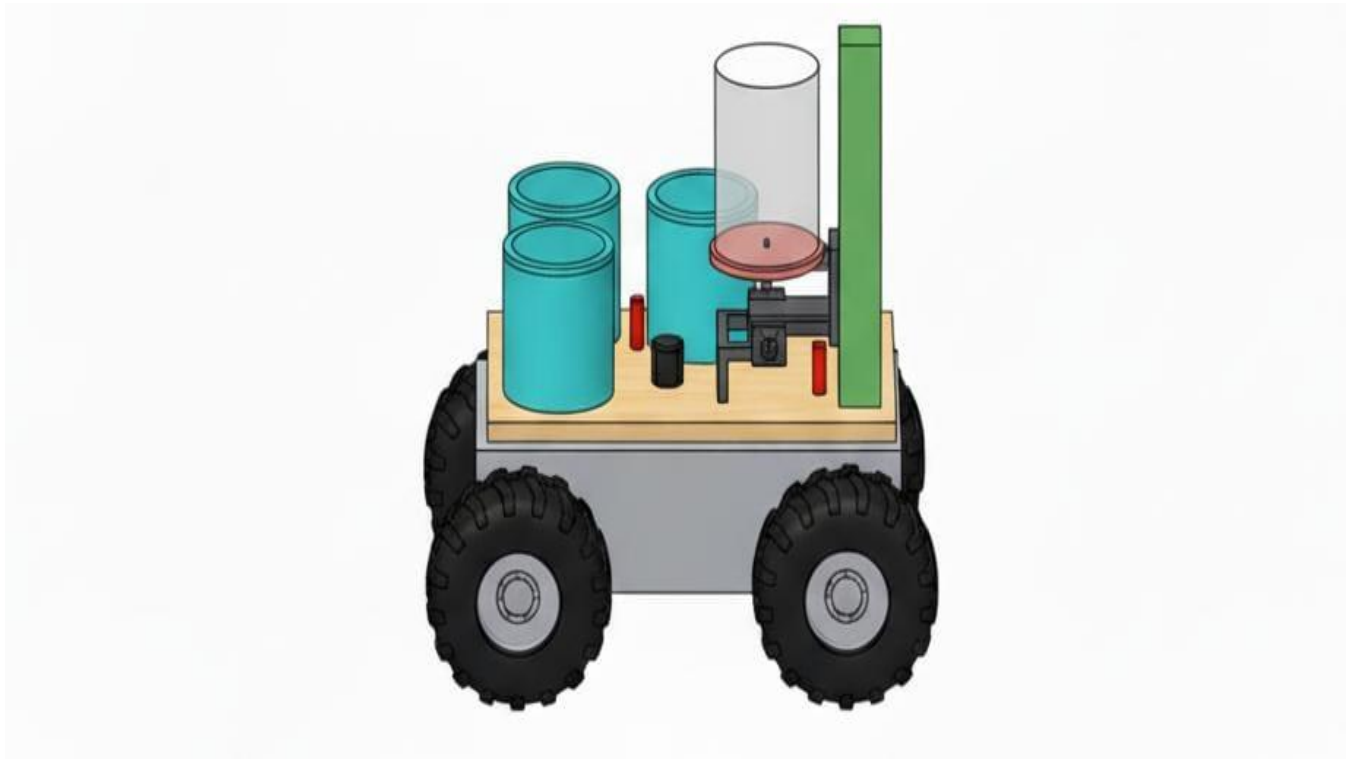


Fig 4.1Side view

42.Front view:

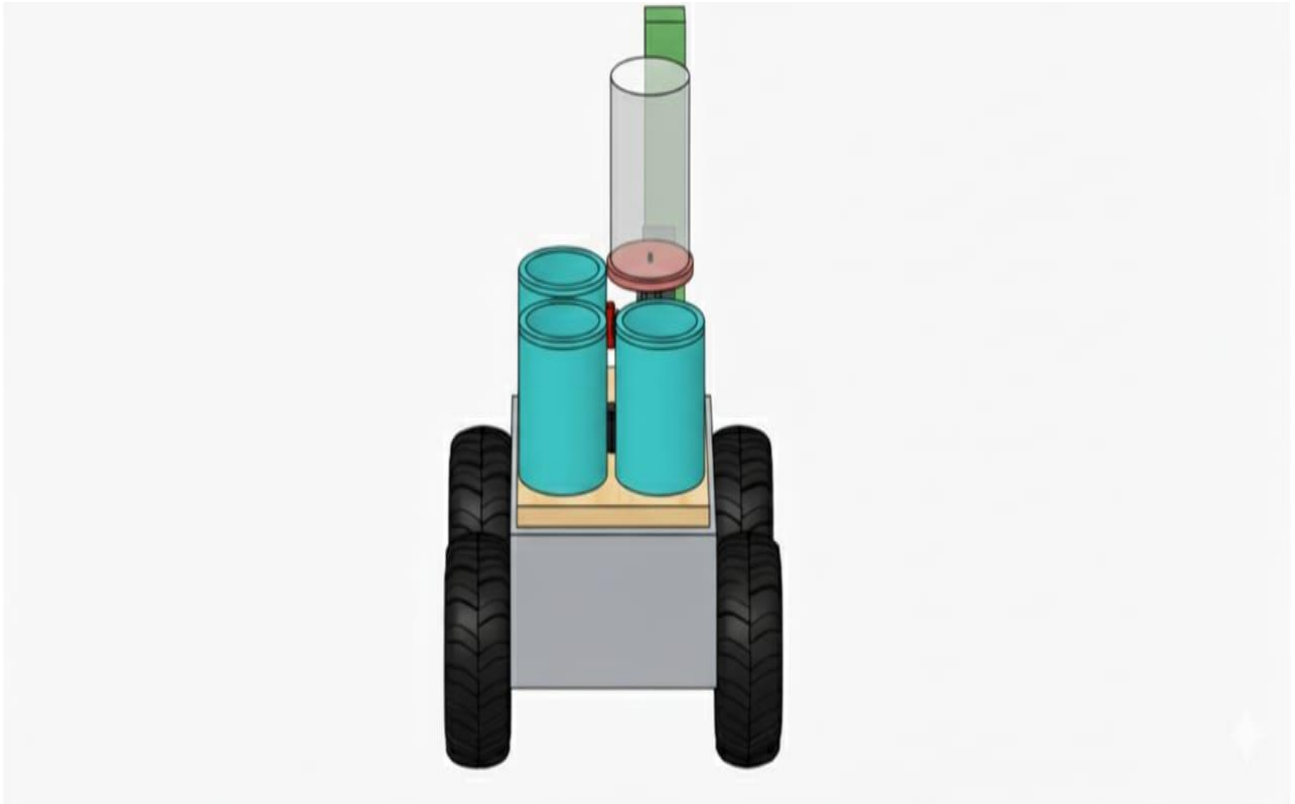


Fig 4.2 Front view

4.3 Top view:

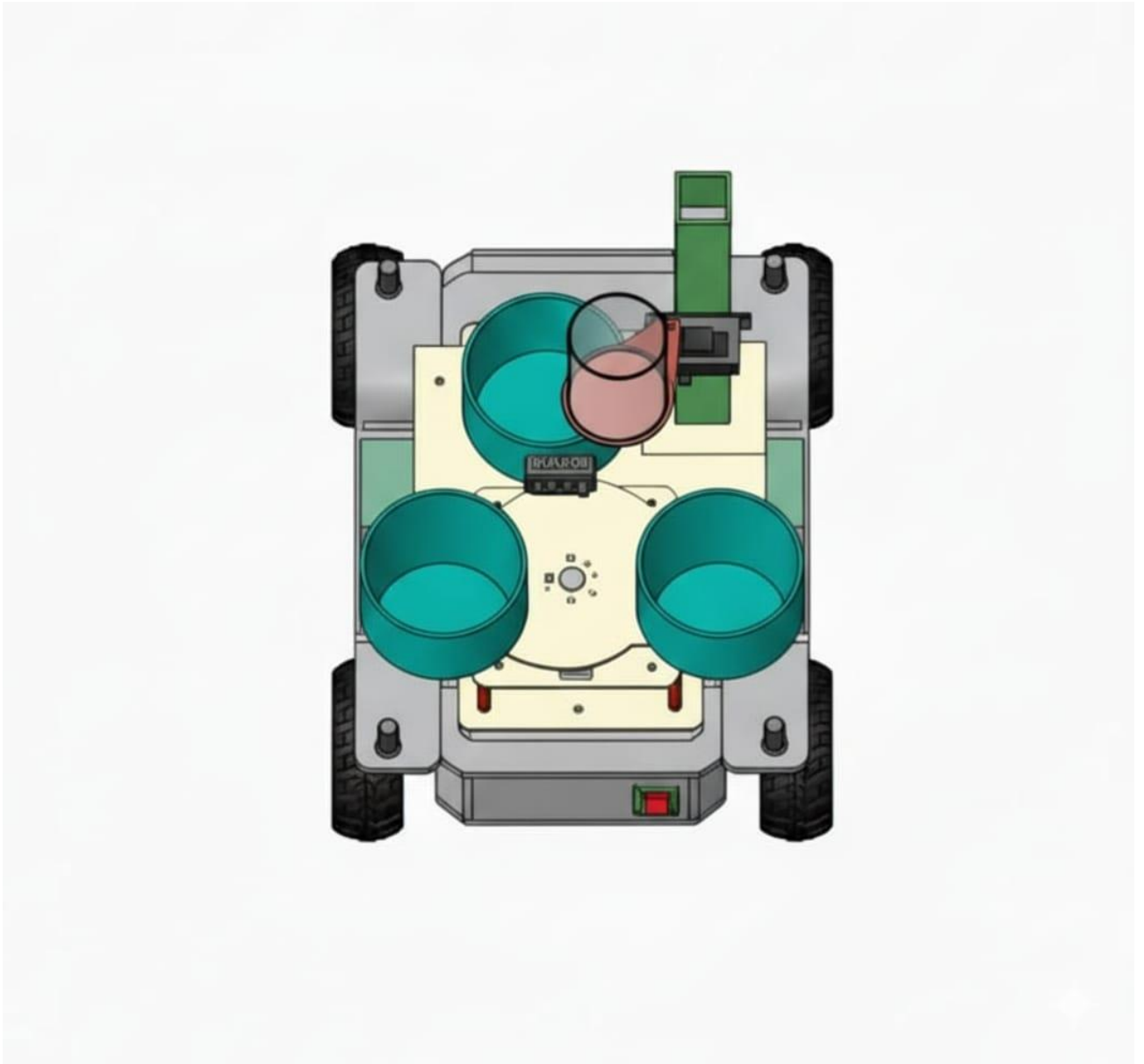


Fig 4.3 Top view

4.4 Isometric view

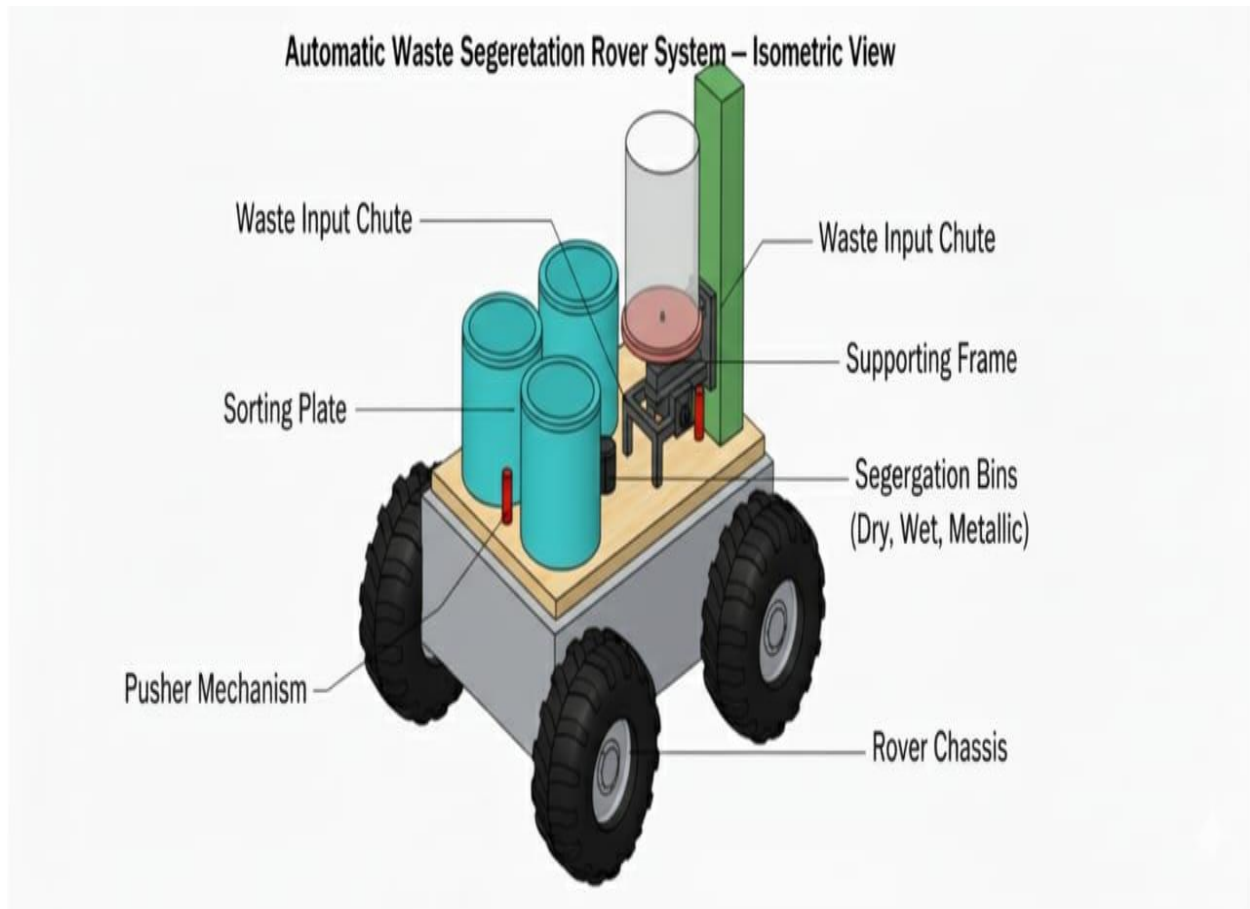


Figure 4.4 shows the isometric view of the proposed work. The part labels are depicted in the figure.

Fig 4.4 Isometric view

4.2 SCHEMATICS AND CIRCUIT DIAGRAM

4.2.1 schematics and circuit diagram

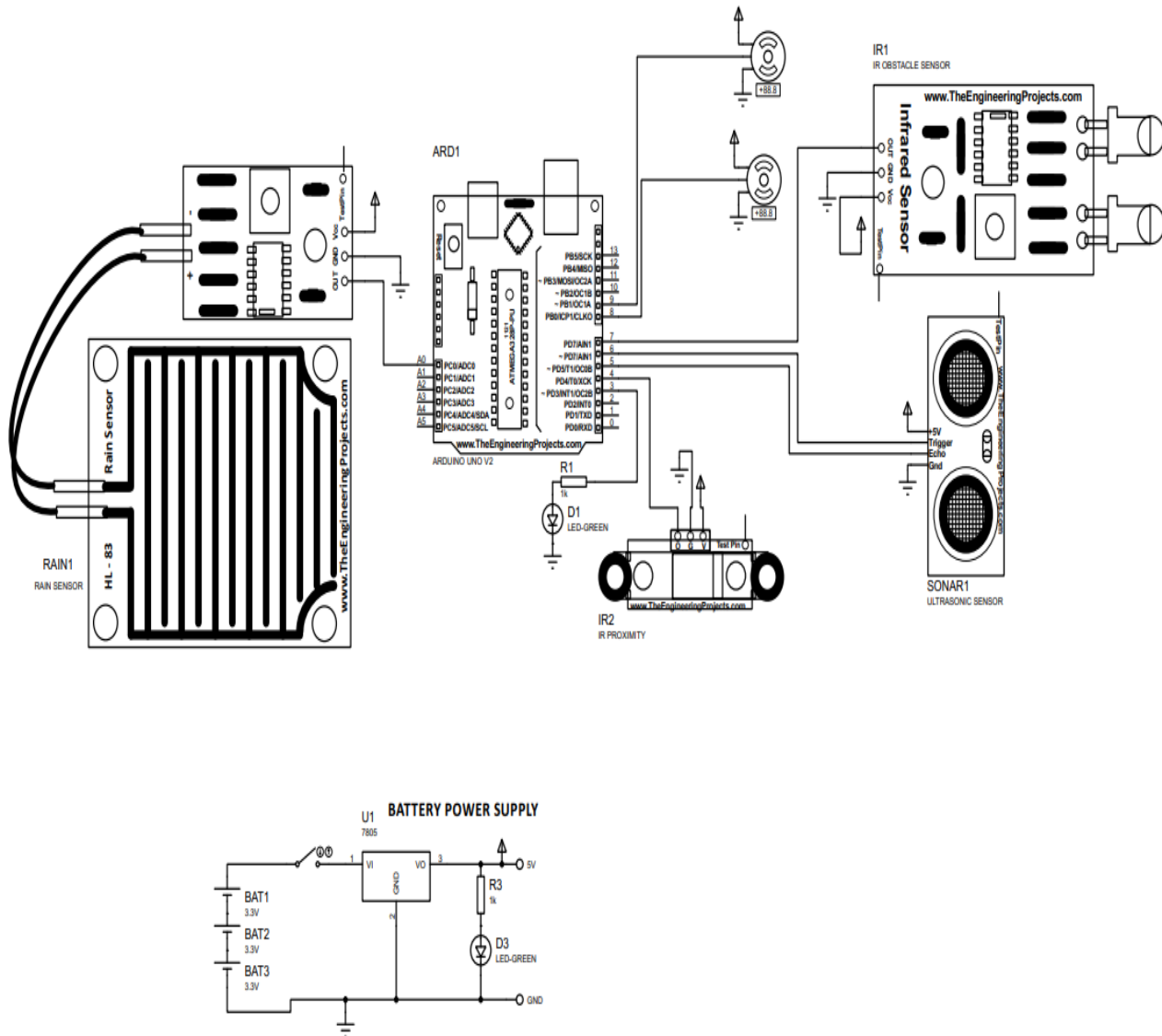


Fig no 4.5 schematics and circuit diagram

4.2.2 SCHEMATIC DIAGRAM OF IR SENSOR

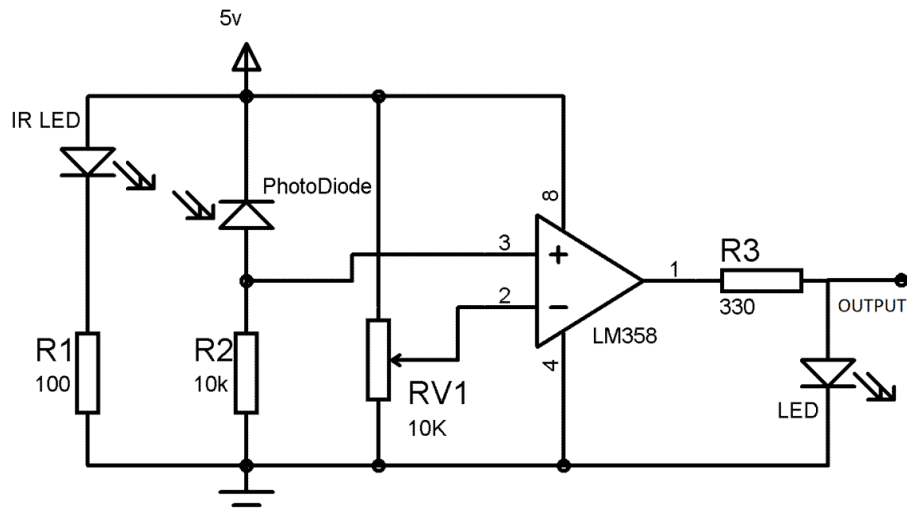


Fig 4.2 IR SENSOR schematic diagram

4.2.3 SCHEMATIC DIAGRAM OF RAIN SENSOR

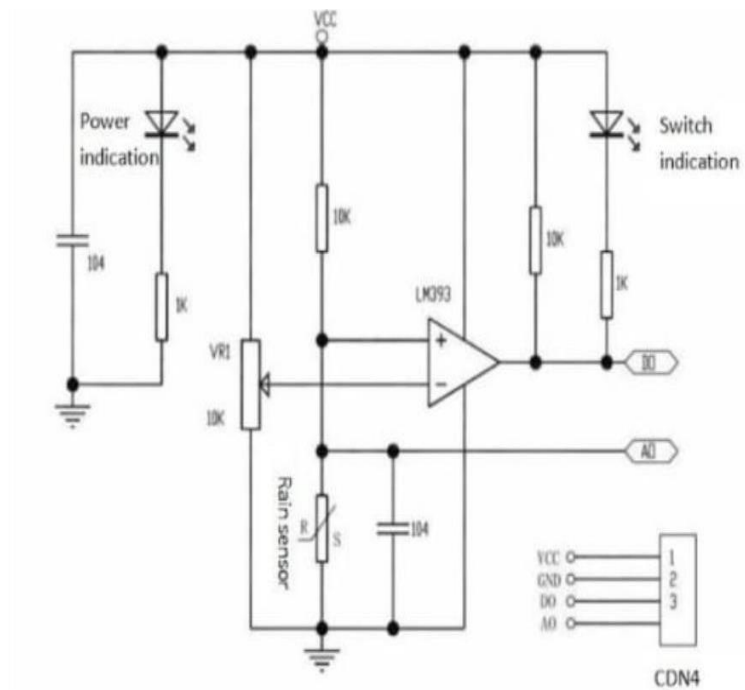


Fig 4.3 Rain sensor schematic diagram

CHAPTER 5

MATERIALS AND METHODS

5.1 COMPONENTS SPECIFICATION:

5.1.1 Arduino Uno Rev3 SMD

The Arduino Uno Rev3 SMD is a microcontroller board based on the ATmega328. It has 14 digital input/output pins (of which 6 can be used as PWM outputs), 6 analog inputs, a 16 MHz ceramic resonator (CSTCE16M0V53-R0), a USB connection, a power jack, an ICSP header, and a reset button. It contains everything needed to support the microcontroller; simply connect it to a computer with a USB cable or power it with a AC-to-DC adapter or battery to get started.

The Uno differs from all preceding boards in that it does not use the FTDI USB-to-serial driver chip.

Additional features coming with the R3 version are:

- ATmega16U2 instead 8U2 as USB-to-Serial converter.
- 1.0 pinout: added SDA and SCL pins for TWI communication placed near to the AREF pin and two other new pins placed near to the RESET pin, the IOREF that allow the shields to adapt to the voltage provided from the board and the second one is a not connected pin, that is reserved for future purposes.
- stronger RESET circuit.

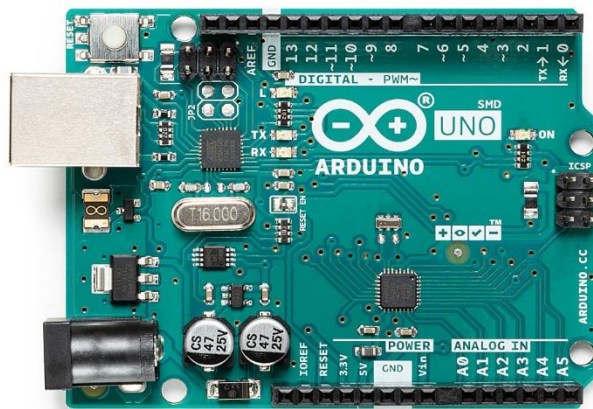


Fig 5.1 Arduino Uno Rev3 SMD

5.1.2 ESP32 MICROCONTROLLER

The ESP32 is a Wi-Fi + Bluetooth-enabled microcontroller that combines high performance, low power consumption, and rich peripheral interfaces. It's the successor to the ESP8266 and is known for its dual-core processor, built-in wireless connectivity, and flexibility in programming (Arduino IDE, MicroPython, etc.).

5.1.3 SENSORS

IR SENSOR : The basic concept of IR(infrared) obstacle detection is to transmit the IR signal(radiation) in a direction and a signal is received at the IR receiver when the IR radiation bounces back from a surface of the object.

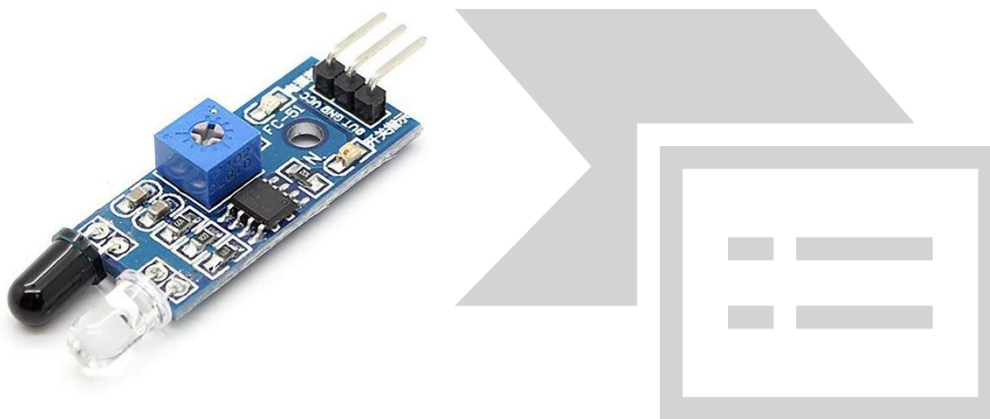


Fig 5.2 IR sensor

RAIN SENSOR : The rain sensor module is an easy tool for rain detection. It can be used as a switch when raindrop falls through the raining board and also for measuring rainfall intensity.

INDUCTIVE PROXIMITY SENSOR : An inductive proximity sensor is a non-contact sensor that detects the presence of metallic objects using electromagnetic principles.

ULTRASONIC SENSOR : An ultrasonic sensor is an electronic device that uses high-frequency sound waves to measure distance, detect objects, and determine specifications.

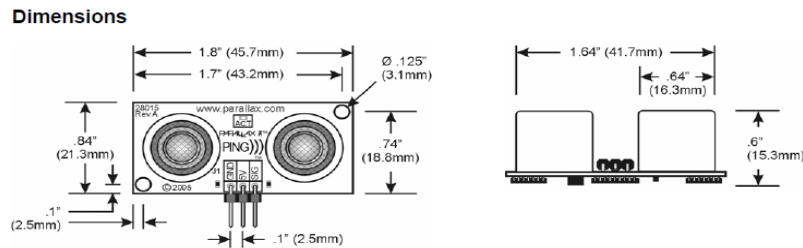


Fig 5.3 Ultrasonic sensor

5.1.4. BATTERY : batteries are used for storage of excess solar energy converted into electrical energy. The only exceptions are isolated sunshine load such as irrigation pumps or drinking water supplies for storage. In fact for small units with output less than one kilowatt. Batteries seem to be the only technically and economically available storage means.

5.1.5 LED : a light-emitting diode (LED) is essentially a PN junction semiconductor diode that emits light when current is applied. By definition, it is a solid-state device that controls current without heated filaments and is therefore very reliable.

5.1.6 MOTORS AND ITS DRIVERS

SERVOMOTOR : A rotary or linear actuator that allows for precise control of angular or linear position, velocity, and acceleration in a mechanical system

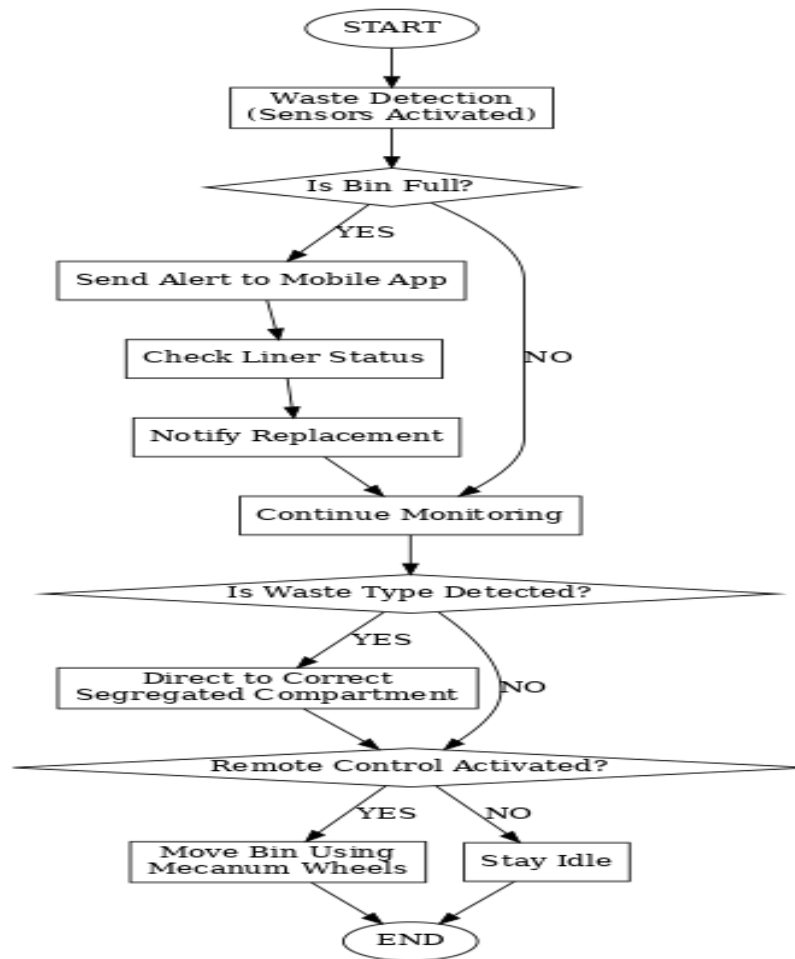


Fig 5.4 Servo motor

GEAR MOTOR : A gear motor is a single, compact unit combining an electric motor with a gearbox or gear reducer

L298N DRIVER : An L298N motor driver is an integrated circuit and associated module used in robotics and automation to control the speed and direction of DC and stepper motors.

5.1.7 Flow chart:



The flowchart details the automated process of a waste segregation system. It begins by detecting waste through sensors, then checks if the bin is full. If full, it alerts a mobile app and manages the liner's status for replacement. Continuous monitoring allows identification of the waste type, directing it to the appropriate compartment. The system also supports remote control: if activated, the bin is moved using mecanum wheels; if not, it stays idle. This process ensures efficient waste management, timely alerts for maintenance, and flexible movement, maximizing both automation and user interaction for smart waste segregation.

Table Chapter 5.2: Specification of Arduino

SPECIFICATION	DETAILS
Core Architecture	AVR (ATmega328P microcontroller)
Operating Voltage	5V
Maximum Clock Speed	16 MHz
Program Memory (Flash)	32 Kb(0.5 KB used by bootloader)
SRAM	2 KB
EEPROM	1 KB
Number of I/O Pins	20 GPIO pins
ADC	10-bit, 6 channels
PWM Channels	6 (Pins 3, 5, 6, 9, 10, 11)
Communication Interfaces	UART, SPI, 12C External Oscillator
Power Consumption	50 mA

Table no 5.1.2 specification of Arduino Uno Rev3 SMD

Table Chapter 5.3: Specification of Esp32 Microcontroller

SPECIFICATION	DETAILS
Core Architecture	Xtensa 32-bit LX6 Dual-Core
Operating Voltage	3.0V – 3.6V
Maximum Clock Speed	240 MHz
Program Memory (Flash)	4 MB external SPI Flash
SRAM	520KB
EEPROM	Not available
Number of I/O Pins	30 GPIO pins
ADC	12-bit, 18 channels
PWM Channels	16 Pins
Communication Interfaces	UART,SPI,12C,Wi-Fi,Bluetooth
Power Consumption	160 – 260 mA

Table no 5.3 specification of ESP32 Microcontroller

Appendix 1

5.3 Programming code:

```
const int rainSensorPin = A0;    // Rain sensor analog pin
const int rainThreshold = 900;   // Adjust according to your sensor readings
const int trigPin = 6;           // Ultrasonic Trigger
const int echoPin = 5;           // Ultrasonic Echo
const int metalSensorPin = 4;     // Metal proximity sensor (digital)
const int greenLedPin = 3;        // Green LED
const int irSensorPin = 4;        // IR sensor digital pin

int rainValue;
int metal = 0; // flag for metal detection

void setup()
{
  Serial.begin(9600);

  servo1.attach(10); // Servo 1 on pin 10
  servo2.attach(11); // Servo 2 on pin 11

  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
  pinMode(metalSensorPin, INPUT);
  pinMode(greenLedPin, OUTPUT);
  pinMode(irSensorPin, INPUT);

  // Initial positions
  servo1.write(90);
  servo2.write(90);
}

// === Loop ===
void loop()
{
  // --- Rain Sensor ---
  rainValue = analogRead(rainSensorPin);
  Serial.print("Rain Value: ");
  Serial.println(rainValue);

  int irState = digitalRead(irSensorPin);
  int metalDetected = digitalRead(metalSensorPin);
```

```

if (rainValue < rainThreshold)
{
  Serial.println("Wet detected...");
  smoothMove(servo1, 90, 0, 15);
  delay(500);
  smoothMove(servo1, 0, 90, 15);
}

if (metalDetected == LOW)
{
  metal = 1;
  Serial.println("METAL DETECTED...");
  smoothMove(servo1, 90, 180, 15);
  delay(500);
  smoothMove(servo1, 180, 90, 15);
}

if (irState == LOW)
{
  Serial.println("DOOR OPENED.....");
  smoothMove(servo2, 90, 0, 15);
  delay(500);
  smoothMove(servo2, 0, 90, 15);
}
else
{
  servo1.write(90);
  servo2.write(90);
  digitalWrite(greenLedPin, LOW);
  Serial.println("IR HIGH - System Stopped");
  metal = 0;
}

long distance = getDistance();
Serial.print("Distance: ");
Serial.println(distance);

if (distance < 5) {
  digitalWrite(greenLedPin, HIGH);
} else {
  digitalWrite(greenLedPin, LOW);
}

```

```

    delay(500);
}

long getDistance()
{
    digitalWrite(trigPin, LOW);
    delayMicroseconds(2);

    digitalWrite(trigPin, HIGH);
    delayMicroseconds(10);
    digitalWrite(trigPin, LOW);

    long duration = pulseIn(echoPin, HIGH);
    long distance = duration * 0.034 / 2; // Convert to cm
    return distance;
}

void smoothMove(Servo &servo, int fromAngle, int toAngle, int stepDelay)
{
    if (fromAngle < toAngle)
    {
        for (int pos = fromAngle; pos <= toAngle; pos++)
        {
            servo.write(pos);
            delay(stepDelay);
        }
    }
    else
    {
        for (int pos = fromAngle; pos >= toAngle; pos--)
        {
            servo.write(pos);
            delay(stepDelay);
        }
    }
}

```

CHAPTER 6

COST ESTIMATION

6.1 ESTIMATION TABLE:

S.No	Parts	Quantity	RS
1	ARDUINO UNO REV3 SMD	1	600
2	ULTRASONIC SENSOR	1	100
3	Rain sensor	1	100
4	PCB Board (15*10)	1	600
5	5V SERVO Motor	2	500
6	ACRYLIC SHEET	1	250
7	JUMPER WIRES(M-F,F-F,M-M)	200	800
8	INDUCTIVE PROXIMITY SENSOR	1	350
9	ROCKER Switch	2	100
10	Battery(5000mAH)	1	1000
11	ESP32 MICROCONTROLLER	1	300
12	Gear Motor	4	500
13	L298N Motor Driver	2	300
14	PVC PIPE (15 CM)	1	20
		Total	5520

Fig no 6.1 estimation table

6.2 ADVANTAGES AND DISADVANTAGES & APPLICATIONS

ADVANTAGES

1. **Efficient Waste Management** – Automates the segregation of wet, dry, and metallic waste, reducing manual effort.
2. **IoT Connectivity** – Enables **real-time monitoring**, remote data access, and analytics for better decision-making.
3. **Environmentally Friendly** – Supports recycling and reduces landfill contribution by proper waste categorization.
4. **Hygienic Process** – Minimizes human contact with waste, lowering health risks.
5. **Smart Data Insights** – IoT sensors can track bin fill levels, waste type statistics, and optimize collection routes.
6. **Scalability** – Can be deployed in households, communities, offices, and smart cities.
7. **Supports SDGs** – Contributes to **SDG 7 (Clean Energy)**, **9 (Innovation)**, **11 (Sustainable Cities)**, **12 (Responsible Consumption)**, and **15 (Life on Land)**.

DISADVANTAGES

1. **High Initial Cost** – IoT modules, sensors, and robotic mechanisms increase setup cost.
2. **Maintenance Requirement** – Sensors and motors need regular calibration and servicing.
3. **Power Dependency** – Requires stable electricity or battery backup to function.
4. **Limited Sorting Categories** – Most prototypes can only segregate into **wet, dry, and metal**; complex segregation (plastic types, e-waste) is harder.
5. **Data Privacy & Connectivity Issues** – IoT systems depend on internet access, which may face **latency, downtime, or security risks**.
6. **Space Requirement** – The system may take more physical space compared to normal bins.

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